a

Data Preparation

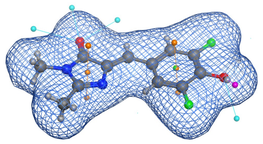
Pretraining Dataset

Ligand-only Molecules ~2.4 M

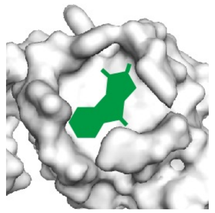
Fine-tuning Dataset

Protein-Ligand Complexes ~127 k

NA-Ligand Complexes 738



Pharmacophore-labelled Ligand Electron Density



Pocket

Ligand Generation

ED Encoder

Pretrain-Only

Constraint (Training Stage Only)

Molecular Generator

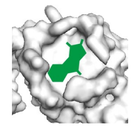
Ligand

Noise

Pocket Encoder

b

Prioritization



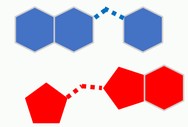
Pocket



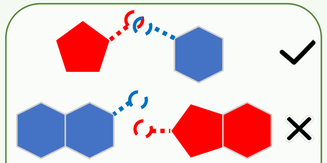


Candidate Ligands

Amplification Pipeline



Fragment Cutting



Fragment Linking

Expanded Library

Virtual Screening

Drug Likeness

Synthetic Accessibility

Docking Score

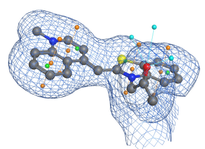
RNAsmol Score

Prioritised candidates

c

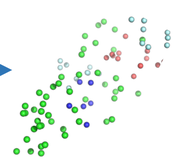
Featurization

Ligand ED Featurization

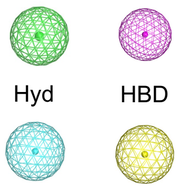


Pharmacophore-labelled Ligand Electron Density

Point Cloud with Pharmacophore Label



Point Cloud

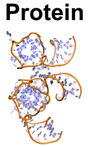


Hyd HBD HBA HBD & HBA

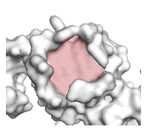
Pocket Featurization



Protein

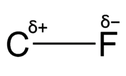


Nucleic Acid



Pocket

Domain-Shared Feature



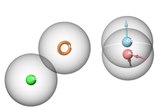
Partial Charge



Hybridization



VdW radii



Pharmacophore

Unified Latent Space : Constraint Transfer & Contrastive Learning

